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# Thermodynamic assessment of the Mn–C system

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ABSTRACT

The manganese–carbon system has been assessed by means of the CALPHAD method. There is a previous description of this system [W. Huang, Scand. J. Metall., 19 (1990) 26–32], but since then a number of high quality experimental investigations have been published. In addition *ab initio* calculations of the enthalpies of formation of the manganese carbides have been performed to support the thermodynamic evaluation. The liquid, $\alpha$-Mn, $\beta$-Mn, $\gamma$-Mn, $\delta$-Mn and $\varepsilon$-Mn phases are described as solution phases and the carbides $Mn_{23}C_6$, $Mn_3C$, $Mn_5C_2$ and $Mn_7C_3$ are described as stoichiometric compounds. A good description of most experimental data, in particular the more recent data, has been achieved. The uncertainties in the thermodynamic properties of the stoichiometric carbides and the liquid are quite small, but fairly large uncertainties remain for the phase equilibria and thermodynamic properties of the solid solution phases.

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## 1. Introduction

The high manganese TRIP ("transformation induced plasticity") and TWIP ("twinning induced plasticity") steels provide great potential in applications for structural components in the automotive industry, owing to their excellent tensile strength-ductility property combination [1]. The development of these materials requires a full understanding of thermodynamic stability of the phases that constitute the ternary Fe–Mn–C system. The study, presented in this paper, is part of a project whose aim is to provide a precise and reliable thermodynamic description for the Fe–Mn–C system. The present work is focused on the assessment of the binary Mn–C system using the CALPHAD approach. This system has been described previously by Huang [2] while liquid of the system has also been assessed by Lee [3,4]. Since then new high quality experimental data on the liquidus [5–9] and the partial Gibbs energy of manganese in various solid state phase equilibria [10] have been published. Data on enthalpy of formation of manganese carbides obtained by *ab initio* calculation in the present work have also been taken into account. The calculated phase diagram of the Mn–C system is shown in Fig. 1.

## 2. Survey of literature information

Experimental thermodynamic and phase equilibrium data of the Mn–C system, available up to 1989, have already been reviewed in detail in a previous evaluation [2]. In the present work, a brief review of these data will be presented. The current understanding of the Mn–C system has evolved gradually and there are still a number of unresolved issues. The first systematic investigation of the Mn–C system was made by Vogel and Döring [11], but there are many discrepancies between their data and subsequently published data since the carbides were not correctly identified at that time. The stability regions of seven solid solution phases ($\alpha$-, $\beta$-, $\gamma$-, $\delta$-Mn and $\alpha$-, $\beta$-, $\gamma$-C) and the position of the liquidus were established by Isobe [12]. Further investigations [13,14] have shown that $\alpha$- and $\gamma$-C do not exist as solid solution phases but as stoichiometric carbides. The $\beta$-C solid solution has been assigned as $\varepsilon$-Mn. The existence and temperature range of stability of five manganese carbides, $Mn_{23}C_6$, $Mn_7C_2$, $Mn_3C$, $Mn_5C_2$ and $Mn_7C_3$, were established by Kuo and Persson [13]. The Mn–C system was re-examined by Benz [14] and a new constitutional diagram was proposed containing five solid solution phases ($\alpha$-, $\beta$-, $\gamma$-, $\delta$-and $\varepsilon$-Mn) and five stoichiometric carbides. The Mn–C phase diagram around the eutectoid decomposition of $\gamma$-Mn and $\varepsilon$-Mn was investigated by Sichen et al. [15]. The temperature and composition of the eutectoid decomposition of $\gamma$-Mn by the reaction $\gamma\text{-Mn} \rightarrow \alpha\text{-Mn} + Mn_{23}C_6$ were estimated to be 1031 K and 9.62 mol% of carbon, respectively. The temperature of the decomposition of $\varepsilon$-Mn to $\gamma$-Mn and $Mn_{23}C_6$ was estimated to be 1260 K, and the equilibrium composition of $\varepsilon$-Mn corresponding to this three-phase reaction was 14.89 mol% of carbon. It should be noted that data on the phase boundary between $\alpha$-Mn and $\gamma$-Mn have been considered unreliable since the number of experimental points is very small and the scatter was significant. The position of the liquidus by Isobe [12] was largely confirmed by Schürman and Geissler [16] and they also reported the peritectic reaction

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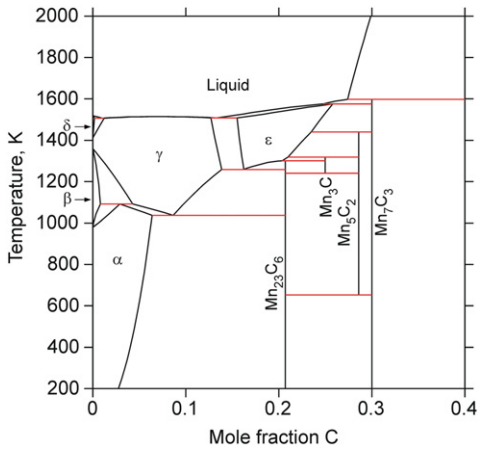


**Fig. 1.** The calculated Mn–C phase diagram.

Liq + $C_{Graphite}$ → $Mn_7C_3$ at 1604 K. However, the liquidus of the $\delta$-Mn phase reported by Isobe [12] and Schürmann and Geissler [16] is not consistent with the heat of melting of $\delta$-Mn and the low solubility of carbon reported by the same authors. Fenstad and Tuset [5] reexamined the high temperature part of the Mn–C phase diagram by differential scanning calorimetry and proposed new liquidus details, such as one local maximum and two eutectics, Liq → $\gamma$-Mn + $\delta$-Mn and Liq → $\gamma$-Mn + $\varepsilon$-Mn at 1505 K and 1501 K, respectively. They have speculated that $Mn_7C_3$ does not form by a peritectic reaction around 1603 K as suggested by Schürmann and Geissler [16], but by a peritectoid reaction around 1561 K. Since there is no strong experimental evidence to support this, the phase relations suggested by Schürmann and Geissler [16] have been chosen in the present work. Consistent values for the solubility of graphite in the liquid phase have been reported by Tanaka [17] and Schenck et al. [18] while higher values have been reported by Turkdogan [19]. However, the new literature data on the solubility of graphite in the liquid phase [6–9] are in fairly good agreement with Turkdogan's data. The temperature of the peritectic reaction involving carbon rich liquid has been determined with high accuracy by means of a radiation thermometer [20].

The manganese and carbon activities in the Mn–C liquid have been reported by Katsnelson et al. [8] at 1628 K and Kim et al. [21] at 1673–1773 K. The obtained activities of manganese are reasonably consistent with Tanaka's data [17] selected by Huang et al. [2]. The partial Gibbs energy of manganese in various solid state phase equilibria has been studied by Zaitsev et al. [10] using Knudsen Effusion Mass Spectrometry (KEMS) in the temperature range 1000–1292 K. The data are in good agreement with the data on the partial Gibbs energy of manganese in various solid state phase equilibria obtained by the EMF technique with a solid electrolyte [22,23] selected by Huang et al. [2]. Zaitsev's results do not support the existence of a carbide at a carbon concentration of 22.2 at.%, referred to as $Mn_7C_2$ carbide by Kuo and Persson [13] and as hexagonal $Mn_{15}C_4$ by Bouchaud and Fruchart [24]. It should also be noted that Zaitsev's results do not support speculations that $Mn_7C_3$ may have two or more structural modifications [24, 25] which are stable over different temperature ranges, or the existence of other carbides [22] richer in carbon than $Mn_7C_3$. A small carbon deficit in $Mn_7C_3$, as reported by Butylenko et al. [26], could be possible.

The low temperature heat capacity [27] in the temperature range from 51 K to 298 K and the high temperature enthalpy increment in the temperature range from 871 K to 1371 K [28] of the $Mn_3C$ phase have been reported. However, current knowledge shows that the measurements do not relate to the $Mn_3C$ phase, but most probably a mixture of $Mn_{23}C_6$ and $Mn_5C_2$. A phase identification was not made in the original investigations.

The standard enthalpies of formation of $Mn_5C_2$ and $Mn_7C_3$ have been determined by high-temperature reaction calorimetry [29] while the standard enthalpy of formation of $Mn_{23}C_6$ has been measured by adiabatic oxygen combustion calorimetry [30].

All relevant experimental phase diagram and thermodynamic data from literature are summarised in Table 1.

## 3. Ab initio calculations

In order to obtain the enthalpies of formation $\Delta H_f$ of the already known ($Mn_3C$, $Mn_5C_2$, $Mn_7C_3$, and $Mn_{23}C_6$) and, additionally, two hypothetical manganese carbides (MnC and $Mn_2C$) at 0 K, we performed calculations based on density-functional theory (DFT). Structure optimisations of the binary bulk phases from first principles were achieved with the Vienna *ab initio* simulation package (VASP) [31,32] using projector-augmented waves (PAW) [33,34] with a kinetic energy cutoff of 500 eV; the crystal orbitals were expanded in plane waves. Exchange and correlation terms were described by the generalised gradient approximation (GGA) functionals of Perdew, Burke, and Ernzerhof [35]. The Brillouin zone integration was performed using the scheme of Monkhorst and Pack [36] using at least 2000 $k$-points · atoms for each unit cell. The partial occupancies were considered using the smearing scheme of Methfessel and Paxton [37] with a $\sigma$ set to 0.2 eV. All results rely on well-converged calculations with respect to the $k$-point mesh and the plane-wave kinetic energy cutoff (500 eV). In order to take into account the magnetic properties of the systems, we computed all structures at the spin-polarised level.

We calculated the total energies of the optimised structures of the binary carbides, $\alpha$-Mn and also diamond, to represent elemental carbon. Because of the well-known problems in properly describing the van der Waals interactions between the layers in graphite using the conventional exchange–correlation functionals [38], we calculated the enthalpies not be referring to graphite but to the covalently bonded diamond, then adding the calorimetrically determined enthalpy difference at 0 K ($H$(diamond) − $H$(graphite) = 2412 J/mol) [39].

The heats of formation have been calculated using:

$$\begin{aligned}\Delta H_f(\mathrm{Mn}_x\mathrm{C}_y) &\approx \Delta E_f(\mathrm{Mn}_x\mathrm{C}_y)\\ &= E(\mathrm{Mn}_x\mathrm{C}_y) - xE(\alpha\text{-Mn})\\ &\quad - y[E(\text{diamond}) - 2.412\ \text{kJ/mol of atoms}]. \qquad (1)\end{aligned}$$

The substitution of $\Delta E$ with $\Delta H$ is possible since the term pV is very small for ground-state structures at relaxed volumes. For the here considered compounds the deviation is between $5\cdot10^{-4}$ (MnC) and $6\cdot10^{-4}$ ($Mn_{23}C_6$) kJ per mole of atoms.

## 4. Thermodynamic models and optimisation of parameters

In the CALPHAD method all available thermodynamic and phase equilibrium data for the system are evaluated simultaneously to obtain one set of model equations for the Gibbs energies of all phases as functions of temperature and composition. Some of these expressions contain adjustable coefficients that are often referred to as model parameters. The optimal values for the model parameters providing the best match between available experimental values and the calculated quantities are usually obtained by a weighted non-linear least square minimisation procedure. This methodology is known as thermodynamic optimisation. From these equations, the thermodynamic properties and the phase diagrams can be calculated. In this work, the thermodynamic optimisation and calculations were carried out using the software package Thermo-Calc [40].

The Gibbs energy of stoichiometric phases and end-members of solutions (see below) are represented as a power series in terms of temperature in the form [41]

$$^{\circ}G(T) = a + bT + cT\ln(T) + dT^2 + eT^3 + fT^{-1} + \sum_n g_n T^n, \qquad (2)$$

**Table 1**
Experimental information for the Mn–C system.

| Data description | Temperature (K) | Experimental technique | Phases involved | Ref. | Use |
|---|---|---|---|---|---|
| Phase diagram | 1498–1923 | DSC, DTA, Chemical analysis | $\gamma$-Mn, $\varepsilon$-Mn, $\delta$-Mn, Liquid | [5] | + |
| C solubility in liquid | 1623–1773 | Chemical analysis | Liquid | [6] | + |
| C solubility in liquid | 1648–1773 | Chemical analysis | Liquid | [7] | + |
| Mn and C activities | 1628 | Gas equilibration (Ar–CO) | Liquid | [8] | + |
| C solubility in liquid | 1623–1723 | Chemical analysis | Liquid | [9] | − |
| Partial Gibbs energy of Mn | 1000–1292 | KEMS | $Mn_{23}C_6$, $Mn_3C$, $Mn_5C_2$, $Mn_7C_3$ | [10] | + |
| Phase diagram | 973–1573 | Thermal and Microscopy analysis | $\alpha$-Mn, $\beta$-Mn, $\gamma$-Mn, $\alpha$-$Mn_3C$, $\beta$-$Mn_3C$, Graphite, Liquid | [11] | − |
| Phase diagram | 923–1623 | Thermal, Magnetic, Microscopy and XRD analysis | $\alpha$-Mn, $\beta$-Mn, $\gamma$-Mn, $\delta$-Mn, $\alpha$-C, $\beta$-C, $\gamma$-C, Graphite, Liquid | [12] | +/−[a] |
| Phase diagram | 963–1323 | XRD analysis | $Mn_{23}C_6$, $Mn_7C_2$, $Mn_3C$, $Mn_5C_2$, $Mn_7C_3$ | [13] | + |
| Phase diagram | 973–1383 | XRD analysis | $\alpha$-Mn, $\beta$-Mn, $\gamma$-Mn, $\varepsilon$-Mn, $\delta$-Mn, $Mn_{23}C_6$, $Mn_{15}C_4$, $Mn_3C$, $Mn_5C_2$, $Mn_7C_3$ | [14] | + |
| Phase diagram | 900–1300 | DTA, EMF | $\alpha$-Mn, $\gamma$-Mn, $Mn_{23}C_6$ | [15] | + |
| Phase diagram | 1473–1623 | Thermal and Microscopy analysis | $\gamma$-Mn, $\varepsilon$-Mn, $\delta$-Mn, $Mn_7C_3$, Graphite, Liquid | [16] | − |
| Mn activities | 1673 | Vapour pressure measurement | Liquid | [17] | − |
| C solubility in liquid | 1643–1873 | Chemical analysis | Liquid | [18] | − |
| C solubility in liquid | 1623–1973 | Chemical analysis | Liquid | [19] | − |
| Melting point | 1604 | Thermal analysis | $Mn_7C_3$, Graphite, Liquid | [20] | + |
| Mn and C activities | 1628–1773 | Gas equilibration (Ar–CO) | Liquid | [21] | + |
| Partial Gibbs energy of Mn | 909–1247 | EMF | $Mn_{23}C_6$, $Mn_3C$, $Mn_5C_2$, $Mn_7C_3$ | [22] | − |
| Partial Gibbs energy of Mn | 1000–1150 | EMF | $Mn_7C_3$ | [23] | − |
| Heat capacity | 51–298 | Adiabatic calorimetry | $Mn_3C$ | [27] | − |
| Enthalpy increment | 871–1371 | High-temperature adiabatic calorimetry | $Mn_3C$ | [28] | − |
| Enthalpy of formation | 298 | High-temperature reaction calorimetry | $Mn_5C_2$, $Mn_7C_3$ | [29] | − |
| Enthalpy of formation | 298 | Adiabatic combustion calorimetry | $Mn_{23}C_6$ | [30] | − |

The column "Use" indicates whether the data were used (+) or not used (−) in assessment.
[a] Data on the peritectoid reaction $\beta$-Mn + $\gamma$-Mn → $\alpha$-Mn, only.

where $a$ to $f$ and $g_n$ are coefficients and $n$ represents a set of integers. Since the enthalpies cannot be defined absolutely, the Gibbs energy is referred to the constant enthalpy values of the so-called Stable Element References, $H^{SER}$, at 298.15 K and 1 bar [41]. The reference states are $\alpha$-Mn (bcc_A12), and Graphite. The Gibbs energy functions of the pure elements are taken from Dinsdale [41].

A substitutional solution model is adopted for the description of the liquid phase. The molar Gibbs energy is given by

$$G^{\mathrm{liq}} = x_{\mathrm{Mn}}{}^{\circ}G^{\mathrm{liq}}_{\mathrm{Mn}} + x_{\mathrm{C}}{}^{\circ}G^{\mathrm{liq}}_{\mathrm{C}} + RT(x_{\mathrm{Mn}}\ln x_{\mathrm{Mn}} + x_{\mathrm{C}}\ln y_{\mathrm{C}}) + {}^{E}G^{\mathrm{liq}}, \quad (3)$$

where $x_i$ represents the mole fraction of the pure element $i$, while ${}^{\circ}G^{\mathrm{liq}}_{\mathrm{Mn}}$ and ${}^{\circ}G^{\mathrm{liq}}_{\mathrm{C}}$ are the Gibbs energy of liquid Mn and C, respectively. The remaining term in Eq. (3), ${}^{E}G^{\mathrm{liq}}$, is the excess Gibbs energy, which can be expressed in terms of interaction parameters ($L$) using a Guggenheim type polynomial series [42] (more commonly known as Redlich–Kister):

$${}^{E}G^{\mathrm{liq}} = x_{\mathrm{Mn}}x_{\mathrm{C}}\sum_{\nu=0}^{n}{}^{\nu}L^{\mathrm{liq}}_{\mathrm{Mn,C}}(x_{\mathrm{Mn}} - x_{\mathrm{C}})^{\nu}. \quad (4)$$

The carbon solution in manganese metal is described by an interstitial solution model with one sublattice occupied by manganese atoms and the other occupied by carbon and vacancies, $(\mathrm{Mn})_1(\mathrm{C, Va})_n$, where $n$ corresponds to the ratio of interstitial sites to normal sites in each structure. The stoichiometry ratio $n$ was accepted as 1 for $\alpha$-Mn (cbcc-A12), $\beta$-Mn (cub-A13) and $\gamma$-Mn (fcc-A1), and as 0.5 and 3 for $\varepsilon$-Mn (hcp-A3) and $\delta$-Mn (bcc-A2), respectively [2]. The molar Gibbs energy of each phase ($\varphi$) is given by

$$G^{\varphi} = y_{\mathrm{Va}}{}^{\circ}G^{\varphi}_{\mathrm{Mn:Va}} + y_{\mathrm{C}}{}^{\circ}G^{\varphi}_{\mathrm{Mn:C}} + nRT(y_{\mathrm{Va}}\ln y_{\mathrm{Va}} + y_{\mathrm{C}}\ln y_{\mathrm{C}}) + {}^{E}G^{\varphi} + {}^{\mathrm{mag}}G^{\varphi}, \quad (5)$$

where $y_i$ represents the site fraction of the species $i$ on the second sublattice. The parameters ${}^{\circ}G^{\varphi}_{\mathrm{Mn:Va}}$ are equal to the Gibbs energy of the pure $\alpha$-Mn, $\beta$-Mn, $\gamma$-Mn, $\delta$-Mn, $\varepsilon$-Mn, whereas ${}^{\circ}G^{\varphi}_{\mathrm{Mn:C}}$ corresponds to the Gibbs energy of the hypothetical compounds, MnC, $Mn_2C$ and $MnC_3$, and controls the carbon solubility in solid manganese. The excess Gibbs energy is given by

$${}^{E}G^{\varphi} = y_{\mathrm{C}}y_{\mathrm{Va}}\sum_{\nu=0}^{n}{}^{\nu}L^{\varphi}_{\mathrm{Mn:C,Va}}(y_{\mathrm{C}} - y_{\mathrm{Va}})^{\nu}, \quad (6)$$

where commas separate species on the same sublattice, and colons separate species on different sublattices. The manganese carbides, $Mn_{23}C_6$, $Mn_3C$, $Mn_5C_2$ and $Mn_7C_3$, are treated as stoichiometric phases. Their Gibbs energies together with the ones of end-members of solutions are described by the following expression:

$$G_{\mathrm{Mn}_n\mathrm{C}_m} = a + bT + n{}^{\circ}G^{\alpha}_{\mathrm{Mn}} + m{}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}}. \quad (7)$$

The coefficients $a$ and $b$, for the manganese carbides, were evaluated using data from Zaitsev et al. [10] while the coefficient $a$, for the end-members of $\gamma$ and $\varepsilon$ solid solution phases, was evaluated using data on enthalpies of formation of manganese carbides obtained by ab initio calculation. The coefficient $a$ of the end-member of the $\alpha$ solid solution phase was set equal to the one of the $\gamma$ phase. The coefficient $b$ for the mentioned end-members together with $a$ and $b$ for other end-members and interaction parameters was obtained by the optimisation procedure. The heat capacity functions of manganese carbides and end-members of solutions were estimated from the heat capacity functions of the elemental components in their standard states ($\alpha$-Mn and Graphite) by applying Neumann–Kopp's rule.

The optimisation procedure was performed using the least-square optimisation module PARROT of Thermo-Calc [40]. PARROT can take into account all kinds of thermodynamic and phase diagram data simultaneously. The program minimises the sum of squared errors between the experimentally determined phase diagram and thermodynamic data and the same data calculated from the current set of coefficients. The relative weight of each experimental data point is adjusted and data, which are inconsistent with the majority of the data points, are excluded during the optimisation procedure.

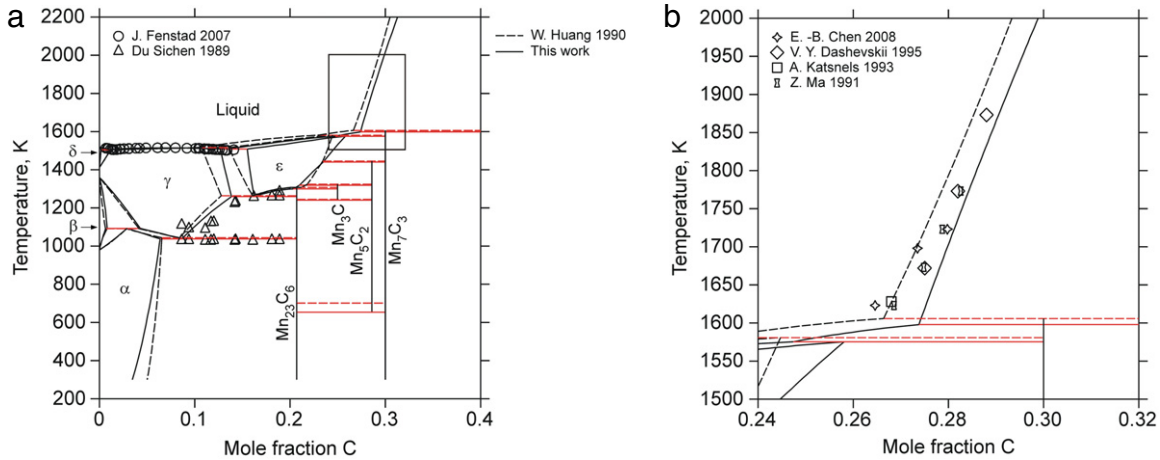


**Fig. 2.** (a) The calculated Mn–C phase diagram (solid lines) compared with the evaluation from Huang [2] (dashed lines) and recent experimental data [5–9,15] (b) is the enlarged view of the square box in (a).

The optimisation was carried out in several steps. First, the parameters of $\alpha$, $\gamma$, and the liquid phases were optimised separately. For the interstitial solid solution phase, $\alpha$, the interaction parameter ${}^{0}L^{\alpha}_{\mathrm{Mn,C}}$ was evaluated based on the measurements of the chemical potential of manganese in the two phase region, $\alpha + \mathrm{Mn_{23}C_6}$ [15], while the same parameter of $\gamma$ was evaluated based on the measurements of chemical potential of manganese in two phase region, $\gamma + \mathrm{Mn_{23}C_6}$, and the carbon solubility in the $\gamma$ phase [15]. It was not possible to simultaneously obtain a good fit of both sets of data when a high positive value of the enthalpy of formation of the metastable MnC end-member of the $\gamma$ phase was used as suggested by the *ab initio* calculation when using only a regular interaction parameter. We, thus, chose to set the enthalpy of formation of metastable MnC to zero, although the *ab initio* calculated value was quite different. Alternatively, it might be possible to use a subregular (in addition to the regular) interaction, but the available experimental data for the $\gamma$ phase are not sufficient to optimism one additional parameter.

For the liquid phase, interaction parameters ${}^{\nu}L^{\mathrm{liq}}_{\mathrm{Mn,C}}$ ($\nu = 0$–2) were determined based on data of the graphite solubility [6–8] and the manganese and carbon activity [21]. All parameters, including the Gibbs functions of stable manganese carbides, were finally optimised simultaneously using the experimental data sets marked in Table 1. The resulting set of parameters for the Mn–C system is presented in Table 2.

**Table 2**
Summary of the thermodynamic parameters describing condensed phases in the Mn–C system referred to stable element reference $H^{\mathrm{SER}}$ ($T = 298.15$ K, $P = 1$ bar). Values are given in SI units (Joule, mole and Kelvin).

| Phase, parameter | Equation |
|---|---|
| Liquid | |
| ${}^{0}L^{\mathrm{liq}}_{\mathrm{Mn,C}} = -179183 + 43.8449T$ | (3) |
| ${}^{1}L^{\mathrm{liq}}_{\mathrm{Mn,C}} = 6313$ | (3) |
| ${}^{2}L^{\mathrm{liq}}_{\mathrm{Mn,C}} = 23281$ | (3) |
| $\alpha$-Mn $(\mathrm{Y})_1(\mathrm{Va, C})_1$ (cbcc-A12) | |
| ${}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn:C}} = {}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + {}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} + 27.46525T$ | (4) |
| ${}^{0}L^{\alpha\text{-Mn}}_{\mathrm{Mn:Va,C}} = -52204$ | (5) |
| $\beta$-Mn $(\mathrm{Mn})_1(\mathrm{Va, C})_1$ (cub-A13) | |
| ${}^{\circ}G^{\beta\text{-Mn}}_{\mathrm{Mn:C}} = {}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + {}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} + 2607$ | (4) |
| ${}^{0}L^{\beta\text{-Mn}}_{\mathrm{Mn:Va,C}} = -10175$ | (5) |
| $\gamma$-Mn $(\mathrm{Mn})_1(\mathrm{Va, C})_1$ (fcc-A1) | |
| ${}^{\circ}G^{\gamma\text{-Mn}}_{\mathrm{Mn:C}} = {}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + {}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} + 13.659T$ | (4) |
| ${}^{0}L^{\gamma\text{-Mn}}_{\mathrm{Mn:Va,C}} = -41333$ | (5) |
| $\delta$-Mn $(\mathrm{Mn})_1(\mathrm{Va, C})_3$ (bcc-A2) | |
| ${}^{\circ}G^{\delta\text{-Mn}}_{\mathrm{Mn:C}} = {}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + 3{}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} + 10000 + 30T$ | (4) |
| $\varepsilon$-Mn $(\mathrm{Mn})_1(\mathrm{Va, C})_{0.5}$ (hcp-A3) | |
| ${}^{\circ}G^{\varepsilon\text{-Mn}}_{\mathrm{Mn:C}} = {}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + 0.5{}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} - 9000 - 1.0651T$ | (4) |
| ${}^{0}L^{\varepsilon\text{-Mn}}_{\mathrm{Mn:Va,C}} = -5006$ | (5) |
| Stoichiometric manganese carbides | |
| ${}^{\circ}G_{\mathrm{Mn_{23}C_6}} = 23{}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + 6{}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} - 310473 + 53.86T$ | |
| ${}^{\circ}G_{\mathrm{Mn_3C}} = 3{}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + {}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} - 39644 + 3.04T$ | |
| ${}^{\circ}G_{\mathrm{Mn_5C_2}} = 5{}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + 2{}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} - 73954 + 6.36T$ | |
| ${}^{\circ}G_{\mathrm{Mn_7C_3}} = 7{}^{\circ}G^{\alpha\text{-Mn}}_{\mathrm{Mn}} + 3{}^{\circ}G^{\mathrm{Graphite}}_{\mathrm{C}} - 106695 + 9.36T$ | |

## 5. Results and discussion

### 5.1. Ab initio calculations

The structures of the already known carbides $\mathrm{Mn_7C_3}$, $\mathrm{Mn_5C_2}$, $\mathrm{Mn_3C}$, and $\mathrm{Mn_{23}C_6}$ were constructed using their experimental structure data [13,43]. The structural relaxations of the four carbides led to only small changes of the lattice parameters and tiny changes of the atomic positions. This confirmed the basic stability (or, at least, metastability) of the structures because an unstable structure would distort upon optimisation. The optimised lattice parameters, as shown in Table 3 together with the magnetic properties and enthalpies of formation, are all 1%–2.5% smaller than the experimentally measured ones. Considering that the calculated values refer to 0 K whereas the measurements are done at room temperature there remains a typical underestimation using this exchange–correlation approximation [44]. All compounds were calculated assuming nonmagnetic, ferromagnetic, and antiferromagnetic orderings. For $\mathrm{Mn_7C_3}$, the calculations confirmed the experimental findings [45] of a nonmagnetic ground state. It is exothermic with a $\Delta H_f = -9.0$ kJ/mol per atom. At first sight, the literature data for $\mathrm{Mn_5C_2}$ is inconsistent, and the phase is labelled as nonmagnetic (above 16 K) [44] or antiferromagnetic [45]. The calculations can neither confirm nor exclude that. The calculated ground state is antiferromagnetic but the energy is only 0.3 kJ/mol per atom more stable than for the nonmagnetic alternative. Therefore, a Néel temperature below 16 K is not implausible. The (open) question of the correct magnetic ground state of $\mathrm{Mn_3C}$ is discussed in detail in a recent paper [46]; the most stable calculated scenario is ferrimagnetic. For $\mathrm{Mn_{23}C_6}$, experimental investigations indicate a magnetic ordering below 108 K, but the exact magnetic structure

**Table 3**
Experimentally reported and *ab initio* calculated lattice parameters, enthalpies of formation and magnetic ordering of the binary manganese carbides.

| Manganese carbide | Space group | Literature [43,44] | | | Present work | | | $\Delta_f H$ (kJ/mol of atoms) | Magnetism |
|---|---|---|---|---|---|---|---|---|---|
| | | *a* (Å) | *b* (Å) | *c* (Å) | *a* (Å) | *b* (Å) | *c* (Å) | | |
| MnC | Fm-3m (No. 225) | – | – | – | 4.09 | – | – | +24.5 | AFM |
| $Mn_2C$ | $P6_3/mmc$ (No. 194) | – | – | – | 2.76 | – | 4.47 | −5.9 | FM |
| $Mn_7C_3$ | Pnma (No. 62) | 4.5430(4) | 6.9544(7) | 11.973(1) | 4.45 | 6.80 | 11.71 | −9.0 | NM |
| $Mn_5C_2$ | C12/c1 (No. 15) | 11.6724(6) | 4.5864(2) | 5.0969(2) | 11.53 | 4.47 | 4.99 | −8.4 | AFM |
| $Mn_3C$ | Pnma (No. 62) | 5.103 | 6.787 | 4.545 | 5.08 | 6.77 | 4.53 | −9.9 | Ferri |
| $Mn_{23}C_6$ | Fm-3m (No. 225) | 10.595(2) | – | – | 10.38 | – | – | −9.3 | Ferri |

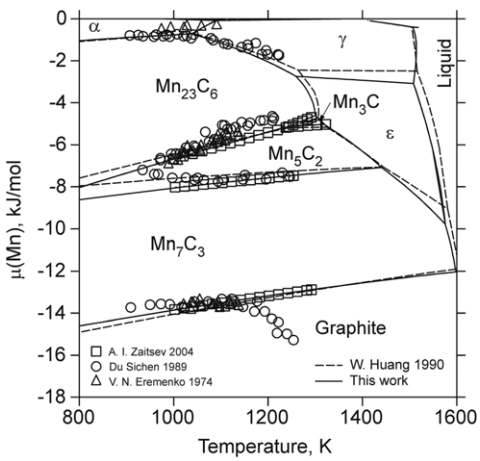


**Fig. 3.** The calculated Mn chemical potential diagram of the Mn–C system (solid lines) compared with the evaluation from Huang [2] (dashed lines) and experimental data [10,15,22,23].

is still unknown [44]. The most stable calculated structure is ferrimagnetic. However, the energy differences between the scenarios are so small that the magnetic structure cannot be solved by *ab initio* methods alone.

The two hypothetical compositions MnC and $Mn_2C$ represent the carbon-rich ends of the $\gamma$ and the $\varepsilon$ phase. Thus, MnC in the sodium chloride type and $Mn_2C$ in the nickel–arsenide type with a 50% occupation of the Ni 2*a* Wyckoff site are constructed. The latter structure type is experimentally known for $MnC_{0.21}$ where carbon occupies the Ni site statistically, too [14]. After adapting the lattice parameters to typical Mn–Mn and Mn–C distances we relaxed the lattice parameters and calculated the enthalpies. In $Mn_2C$ the manganese–carbon distances are 1.95 Å and thus comparable with those in $Mn_3C$ (1.98–2.01 Å), $Mn_5C_2$ (1.99 Å), and $Mn_7C_3$ (1.95 Å). However, the metal–metal distances arrive at 2.753 Å, thus being larger than the ones in the experimentally known carbides (2.37–2.53 Å). The most stable $Mn_2C$ has a ferromagnetic ordering of the Mn atoms with a magnetic saturation moment of $\mu = 2.1\ \mu_B$/Mn atom. NiAs-type $Mn_2C$ is, although unknown, an exothermic compound. In contrast to all other manganese carbides MnC is the only unstable compound. It is antiferromagnetic with a magnetic moment of $\mu = 2.0\ \mu_B$/Mn atom and an enthalpy of formation of $\Delta H_f = 24.5$ kJ/mol of atoms. An indication for this destabilisation is seen in the long bonding distances in the structure. Both the Mn–C (2.05 Å) and the Mn–Mn distances (2.89 Å) are longer than in all other calculated structures. The metal–metal distance is at least 14% longer in this "close packing" than in all known manganese carbides.

### 5.2. Thermodynamic optimisation

The phase and stability diagrams as calculated according to the present evaluation (solid lines) and according to the previous evaluation (dashed lines) along with the available experimental data (points) are shown in Figs. 2 and 3, respectively. The present thermodynamic description, in comparison to previous assessment, yields the liquidus line shifted towards slightly lower temperatures and slightly higher values for the solubility of graphite in the liquid phase. These differences are caused by the use of the new experimental data on the position of the liquidus and solubility of graphite in the liquid phase.

The results of the present and the previous work together with the experimental data, characterising the invariant equilibria occurring in the Mn–C system, are given in Table 4. The temperatures of the all invariant equilibria with the exception of temperature of peritectoid formation of the $Mn_5C_2$ phase are reproduced within experimental error.

The calculated activity coefficients of manganese and carbon in Mn–C melt at temperature ranging from 1628 to 1773 K compared with selected experimental data are shown in Fig. 4(a) and (b),

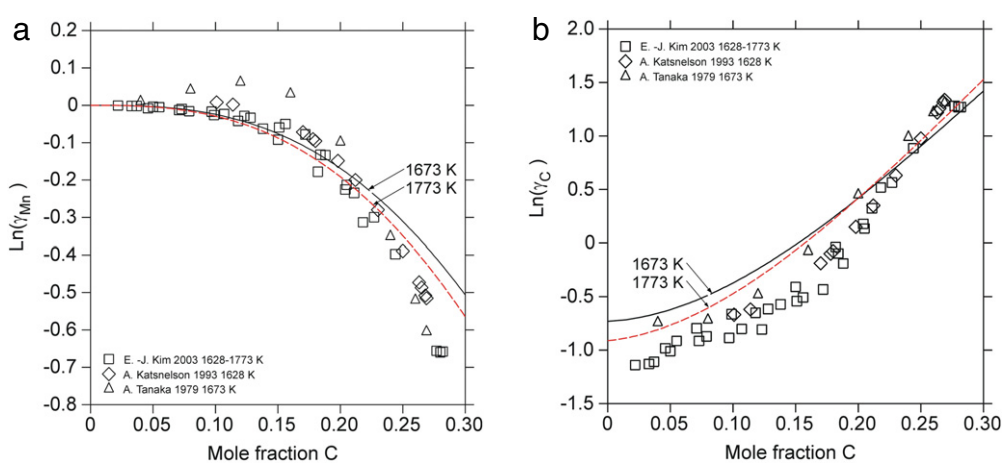


**Fig. 4.** The calculated activity coefficients of manganese (a) and carbon (b) in Mn–C melt at 1673 K (solid lines) and 1773 K (dashed lines) compared with experimental data [8,17,21].

**Table 4**
Invariant reactions in the Mn–C system.

| Invariant reaction | Type | Temperature (K) | | | Mass % of C | | |
|---|---|---|---|---|---|---|---|
| | | Present work | Previous assess. [2] | Literature | Present work | Previous assess. [2] | Literature |
| $L^a \rightarrow \delta\text{-Mn} + \gamma\text{-Mn}$ | Eutectic | 1507 | 1504 | 1506 [5]<br>1509 [16] | 0.22 | 0.25 | 0.31<br>0.90 |
| $L^a \rightarrow \varepsilon\text{-Mn} + \gamma\text{-Mn}$ | Eutectic | 1508 | 1516[b] | 1502 [5]<br>1533[b] [16] | 3.24 | 2.42[b] | 3.25<br>2.50[b] |
| $L^a + C_{Graphite} \rightarrow Mn_7C_3$ | Peritectic | 1598 | 1606 | 1604 [20]<br>1608 [16] | 7.62 | 7.35 | –<br>8.00 |
| $L + Mn_7C_3 \rightarrow \varepsilon\text{-Mn}^a$ | Peritectoid | 1575 | 1581 | 1573 [16] | 6.7 | 6.61 | 6.00 |
| $\varepsilon\text{-Mn}^a + Mn_7C_3 \rightarrow Mn_5C_2$ | Peritectoid | 1440 | 1444 | 1363 [14]<br>1323 [13] | 6.28 | 6.27 | – |
| $\varepsilon\text{-Mn}^a + Mn_5C_2 \rightarrow Mn_3C$ | Peritectoid | 1319 | 1325 | 1323 [14]<br>1323 [13] | 5.49 | 5.76 | – |
| $\varepsilon\text{-Mn}^a + Mn_3C \rightarrow Mn_{23}C_6$ | Peritectoid | 1300 | 1307 | 1293 [14]<br>1298 [13] | 5.28 | 5.55 | – |
| $\varepsilon\text{-Mn}^a \rightarrow \gamma\text{-Mn} + Mn_{23}C_6$ | Eutectoid | 1259 | 1263 | 1260 [15]<br>1263 [14] | 4.06 | 4.02 | 3.68 |
| $Mn_3C \rightarrow Mn_5C_2 + Mn_{23}C_6$ | Eutectoid | 1241 | 1244 | 1243 [14]<br>1240 [10] | | | – |
| $\beta\text{-Mn} + \gamma\text{-Mn} \rightarrow \alpha\text{-Mn}^a$ | Peritectoid | 1092 | 1091 | 1130 [12]<br>1113 [11] | 0.65 | 0.63 | 0.9<br>0.8 |
| $\gamma\text{-Mn}^a \rightarrow \alpha\text{-Mn} + Mn_{23}C_6$ | Eutectoid | 1038 | 1043 | 1031 [15]<br>1163 [14] | 2.03 | 1.97 | 1.84 |
| $Mn_5C_2 \rightarrow Mn_{23}C_6 + Mn_7C_3$ | Eutectoid | 653 | 701 | – | | | – |

[a] Indicates phases, for which the mass % of carbon is listed.
[b] The temperature and the composition are referred to peritectic reaction $L^a + \varepsilon\text{-Mn} \rightarrow \gamma\text{-Mn}$.

**Table 5**
Enthalpies of formation of the binary manganese carbides.

| Manganese carbide | Space group | Prototype | $\Delta_f H^o_{298}$ (kJ/mol of atoms) | | | |
|---|---|---|---|---|---|---|
| | | | Selected experimental values | *Ab initio* (0 K) | Previous assessment [2] | Present work |
| $MnC^a$ | Fm-3m (No. 225) | NaCl | | 24.5 | 0.251 | 0 |
| $Mn_2C^b$ | $P6_3/mmc$ (No. 194) | NiAs | | −5.9 | −11.1 | −6.0 |
| $Mn_7C_3$ | Pnma (No. 62) | $Cr_7C_3$ | −10.7 [10]<br>−9.1 [29] | −9.0 | −11.2 | −10.7 |
| $Mn_5C_2$ | C12/c1 (No. 15) | $Mn_5C_2$ | −10.6 [10]<br>−8.9 [29] | −8.4 | −11.0 | −10.6 |
| $Mn_3C$ | Pnma (No. 62) | $Fe_3C$ | −9.9 [10] | −9.9 | −10.1 | −9.9 |
| $Mn_{23}C_6$ | Fm-3m (No. 225) | $Cr_{23}C_6$ | −10.7 [10]<br>−8.5 [30] | −9.3 | −10.6 | −10.7 |

[a] The carbon rich end-member of the $\gamma$ phase.
[b] The carbon rich end-member of the $\varepsilon$ phase.

respectively. The agreement between calculated and measured values is satisfactory.

The enthalpies of formation of manganese carbides at 298.15 K as calculated according to the present evaluation and the previous optimisation together with the selected experimental and *ab initio* (at 0 K) data are given in Table 5. The present optimisation yields slightly less negative values than the previous assessment [2], except for the $\varepsilon$ phase where a significant difference could be observed. For all stable carbides, the *ab initio* calculation is very close to the experimental and optimised values, with a maximum difference of 2.2 kJ/mol of atoms for $Mn_5C_2$. This gives us confidence that the *ab initio* results are reliable. For the $\varepsilon$ phase this was used to select quite a different value for the $Mn_2C$ end point compared to the previous assessment [2]. For the MnC end point of the $\gamma$ phase we are confident that the *ab initio* value is reasonable as well. However, we could not use this value since it would have required an additional sub-regular interaction parameter, which could not be optimised at present, because there were not sufficient experimental data available for the stable region of the $\gamma$ phase.

It should be noted that problems in the previous thermodynamic descriptions of the Mn–C system, such as stability of the $\varepsilon$ phase just below 298.15 K and an inverse miscibility gap in the liquid phase on the C-rich side, have been resolved in the present description.

## 6. Conclusions

A consistent thermodynamic description of the entire Mn–C system was obtained by considering the *ab initio* calculation values and critically assessed literature data. In comparison to the previous modeling, an improvement has been achieved. However, it should be noted that substantial uncertainties remain

concerning the solid state phase equilibria of the solid solution phases.

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## Appendix. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.calphad.2010.05.002.

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